Project Design Phase-II

Technology Stack (Architecture and Stack)

Team Id	SBAP0030975
Project Name	Visualizing housing market trends: Analysis of sales prices feature using tableau

Technical Architecture

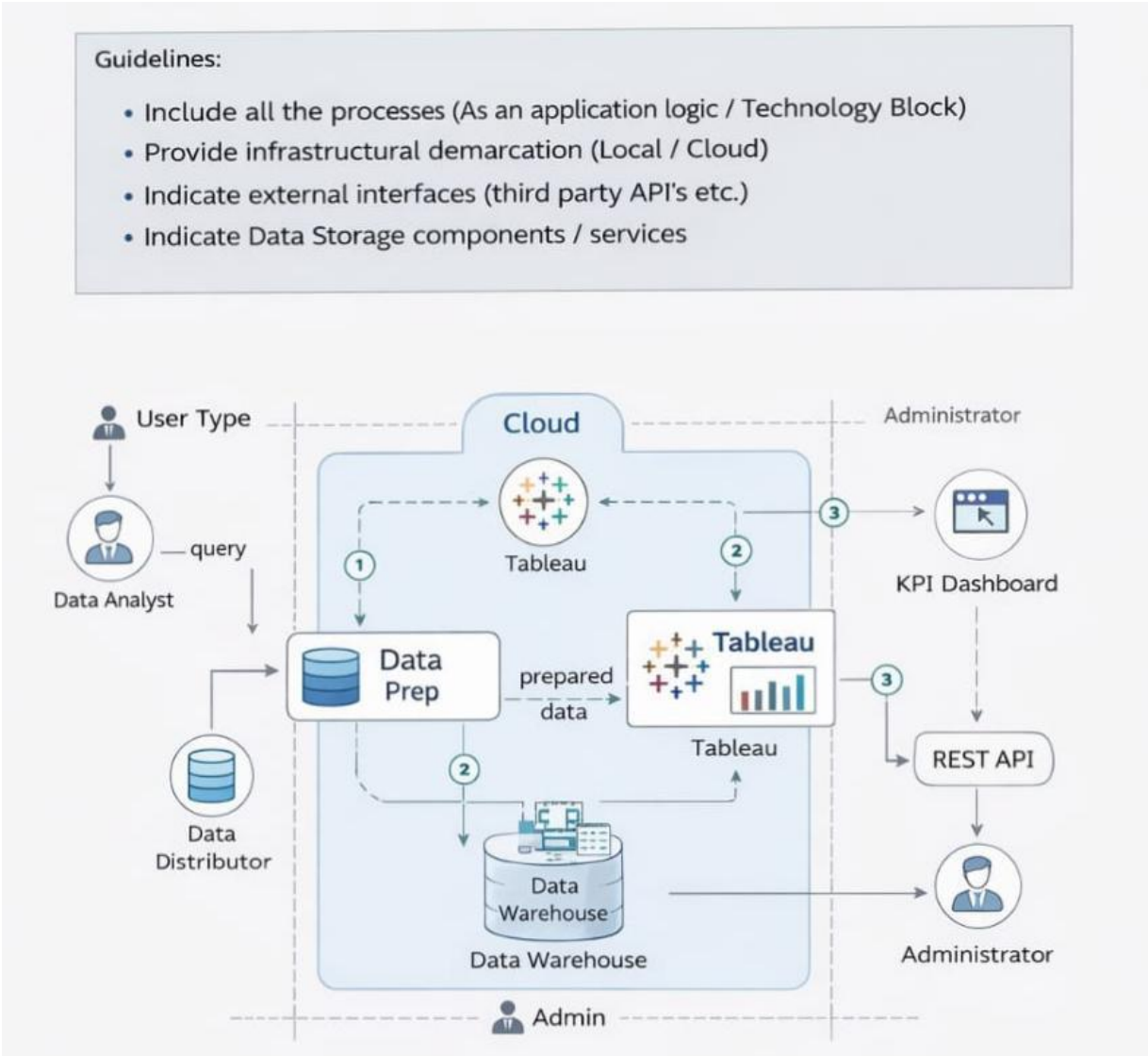

Table -1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards and visual reports for analyzing housing prices and property features.	Tableau Desktop, Tableau Server / Tableau Cloud
2	Application Logic-1	Data preparation, cleaning, transformation, and aggregation processes.	Tableau Prep Builder
3	Application Logic-2	Data modeling, calculated fields (price per sqft, growth rate), KPIs creation.	Tableau Calculated Fields, Tableau Data Model
4	Application Logic-3	Trend analysis, feature comparison, forecasting and statistical analysis.	Tableau Analytics, Trend Lines, Forecasting
5	Database	Storage of raw housing datasets (CSV/Excel/Local DB).	MySQL / PostgreSQL / Excel / CSV
6	Cloud Database/Data Warehouse	Centralized structured storage for integrated housing data.	Amazon Redshift / Google BigQuery / Snowflake
7	File Storage	Storage of raw and processed housing datasets and extracts (.hyper files).	Local File System / Cloud Storage (AWS S3, Google Cloud Storage)
8	External API-1	Fetch real-time housing market data or property listings (if applicable).	Real Estate APIs (e.g., Zillow API, RapidAPI Housing APIs)
9	External API-2	Geographic data for mapping and regional price visualization.	Google Maps API / OpenStreetMap API
10	Machine Learning Model	Predict housing prices and identify future market trends.	Python (Scikit-learn) / Tableau Integration with R or Python
11	Infrastructure	Deployment of dashboards and scheduling refresh of data.	Tableau Server (On-Premise) / Tableau Cloud

Table -2: Application characteristics

S.No	Characteristics	Description	Technology
1	Open source -Fremeworks	List the open-source frameworks used for developing the housing market visualization system (frontend, backend, visualization libraries).	React.js, Angular, Django, Flask, Node.js, D3.js, Chart.js
2	Security Implements	Measures to secure housing data and user access (authentication, encryption).	SSL/TLS, OAuth 2.0, JWT, SHA-256
3	Scalable Architecture	Describe how the system handles increasing housing data volume and users (modular design, microservices, cloud deployment).	Microservices, Docker, Kubernetes, AWS, Azure
4	Availability	Ensure system availability for continuous access to housing trend dashboards (load balancing, cloud hosting, redundancy).	AWS EC2, Load Balancer, Google Cloud, Azure
5	Performance	Design considerations for fast data processing and visualization (caching, optimized queries, CDN usage).	Redis Cache, CDN, SQL Optimization